Lab 3: Assessing the Network with Common Security Tools

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IST 476: Applied Information Security

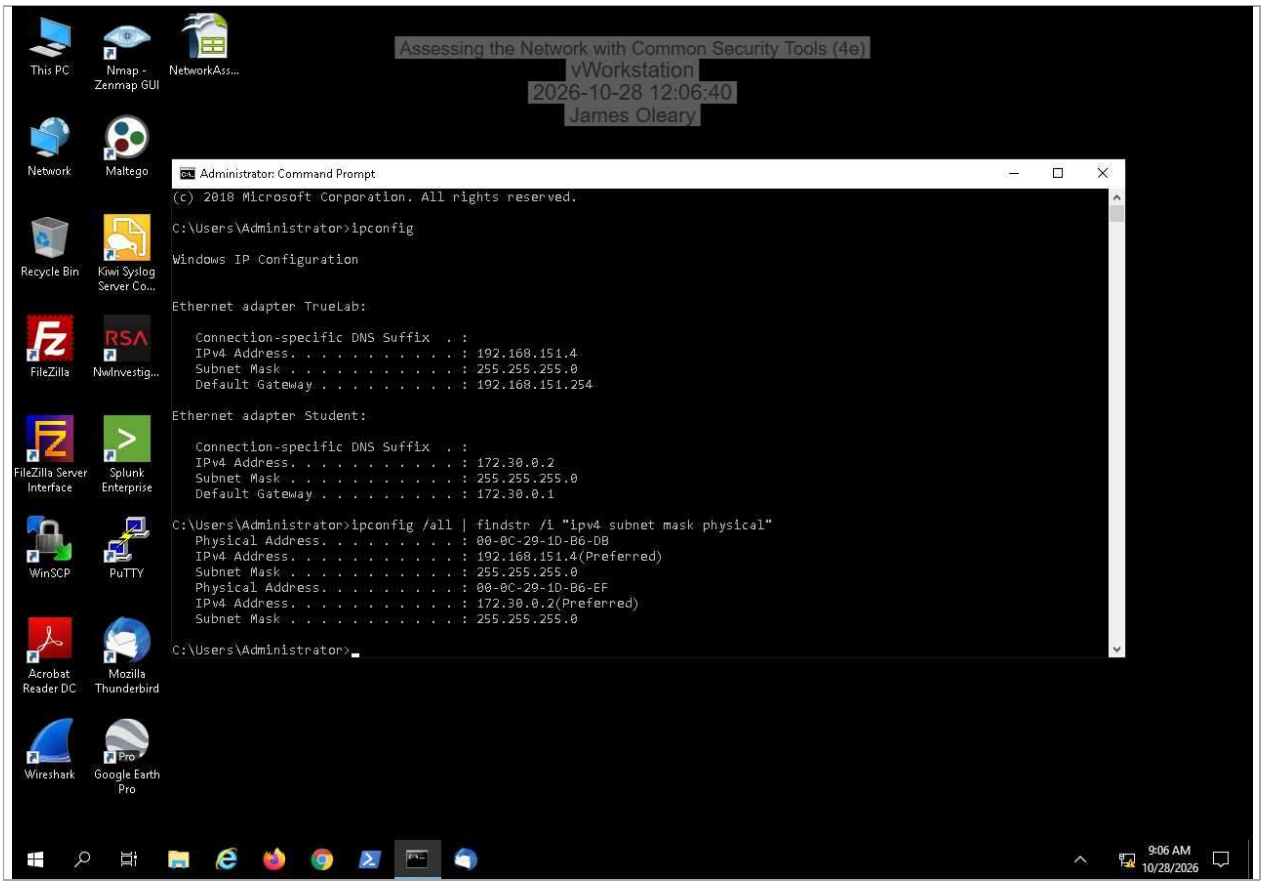
Dr. Joon S. Park

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Section 1, Part 1: Exploring the LAN with Basic Network Utilities

Figure 1

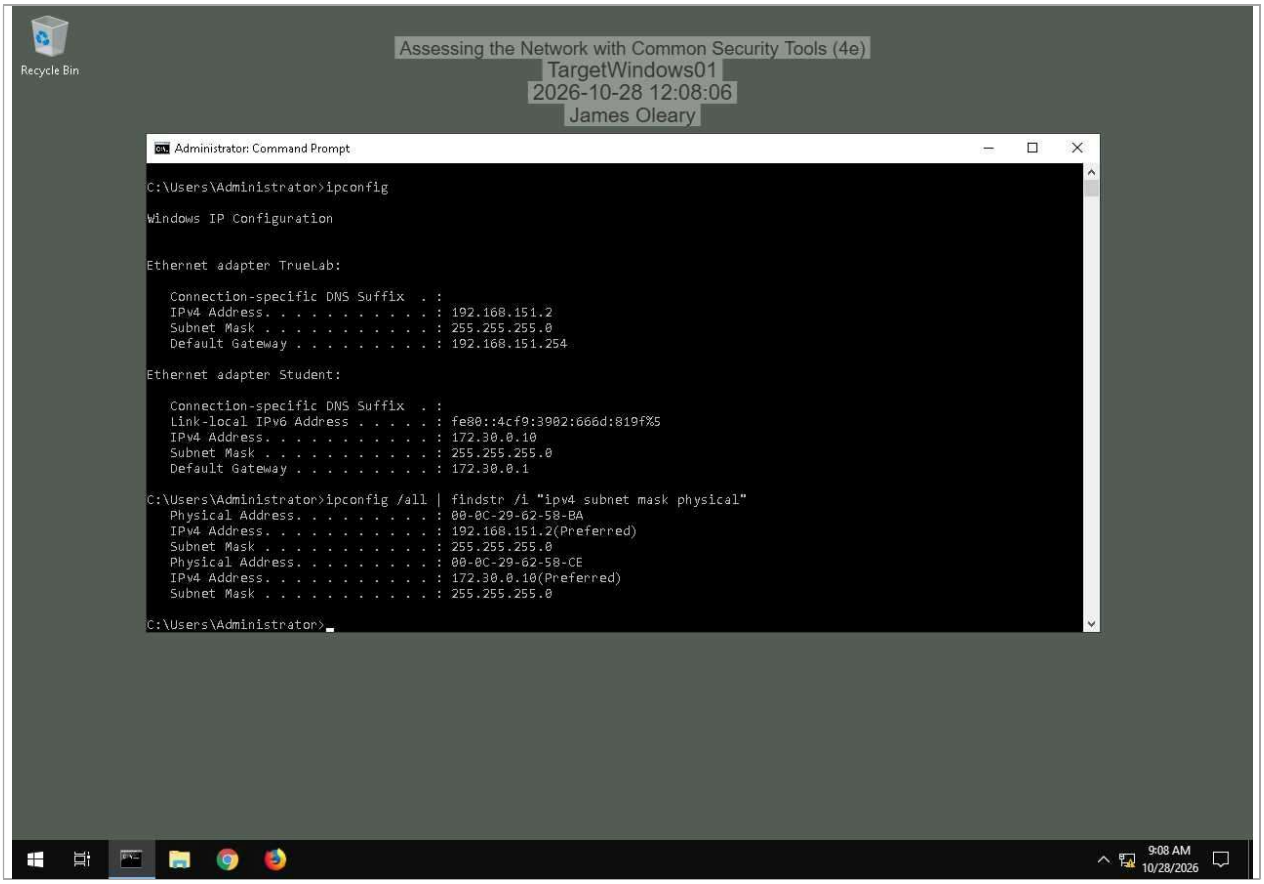
Required Screenshot 1: ipconfig Results for the Student Adapter on vWorkstation



Note. vWorkstation's Student adapter is configured with IPv4 address 172.30.0.2, subnet mask 255.255.255.0, default gateway 172.30.0.1, and MAC address 00-0C-29-1D-B6-EF.

Figure 2

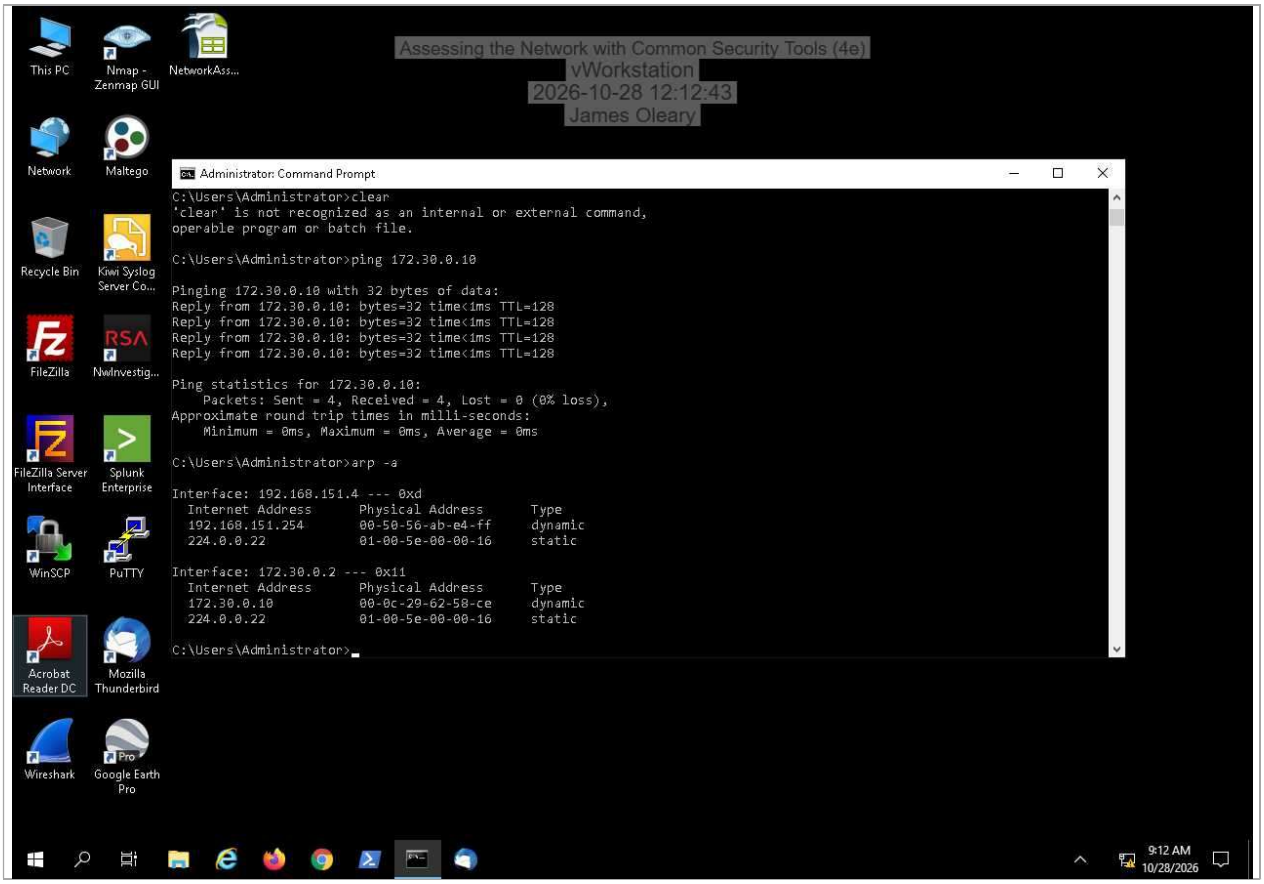
Required Screenshot 2: ipconfig Results for the Student Adapter on TargetWindows01



Note. TargetWindows01's Student adapter is configured with IPv4 address 172.30.0.10, subnet mask 255.255.255.0, default gateway 172.30.0.1, and MAC address 00-0C-29-62-58-CE.

Figure 3

Required Screenshot 3: Updated ARP Cache on vWorkstation After Pinging TargetWindows01

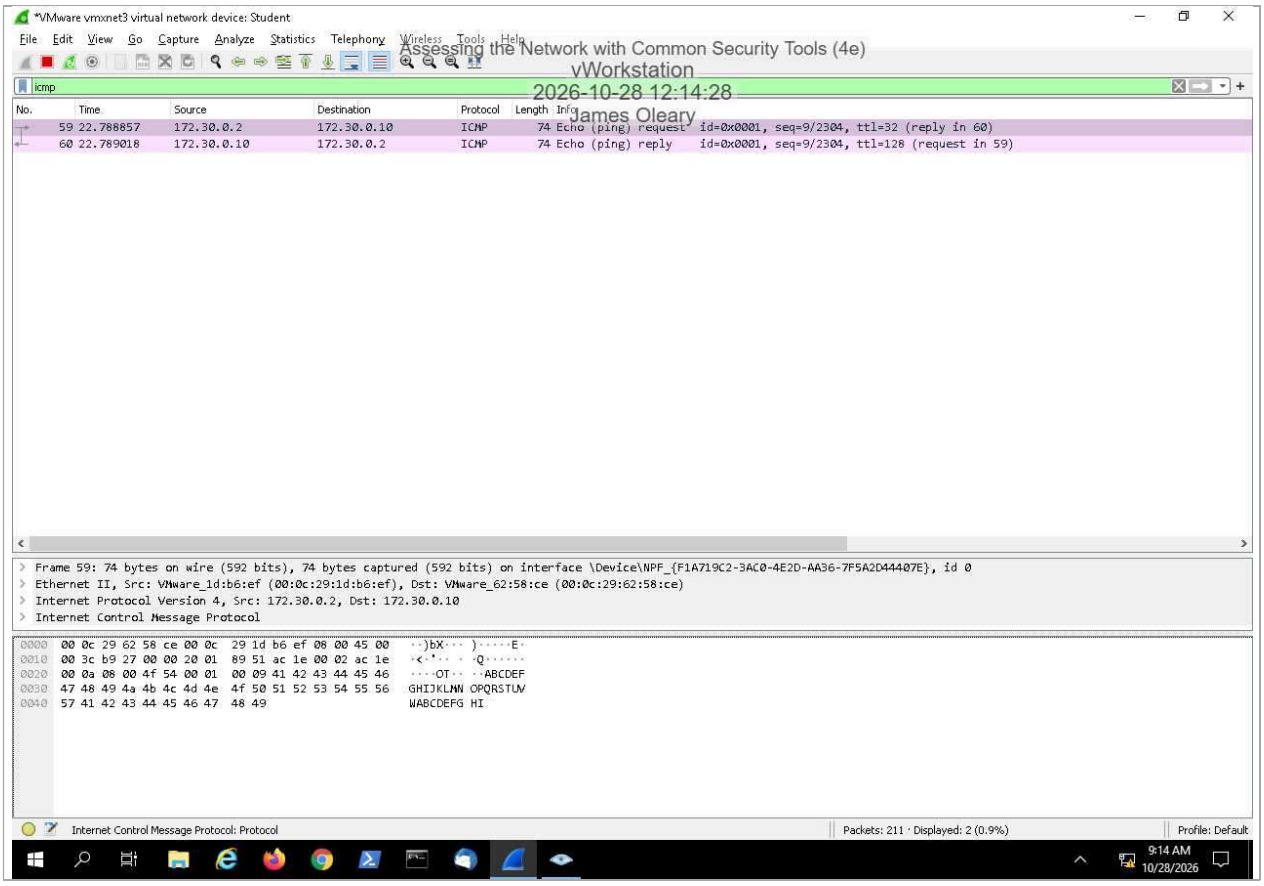


Note. After four successful echo replies from 172.30.0.10, vWorkstation's ARP cache holds a new dynamic entry for 00-0c-29-62-58-ce, matching TargetWindows01's physical address in Figure 2.

Section 1, Part 2: Advanced LAN Analysis with Wireshark and Nmap

Figure 4

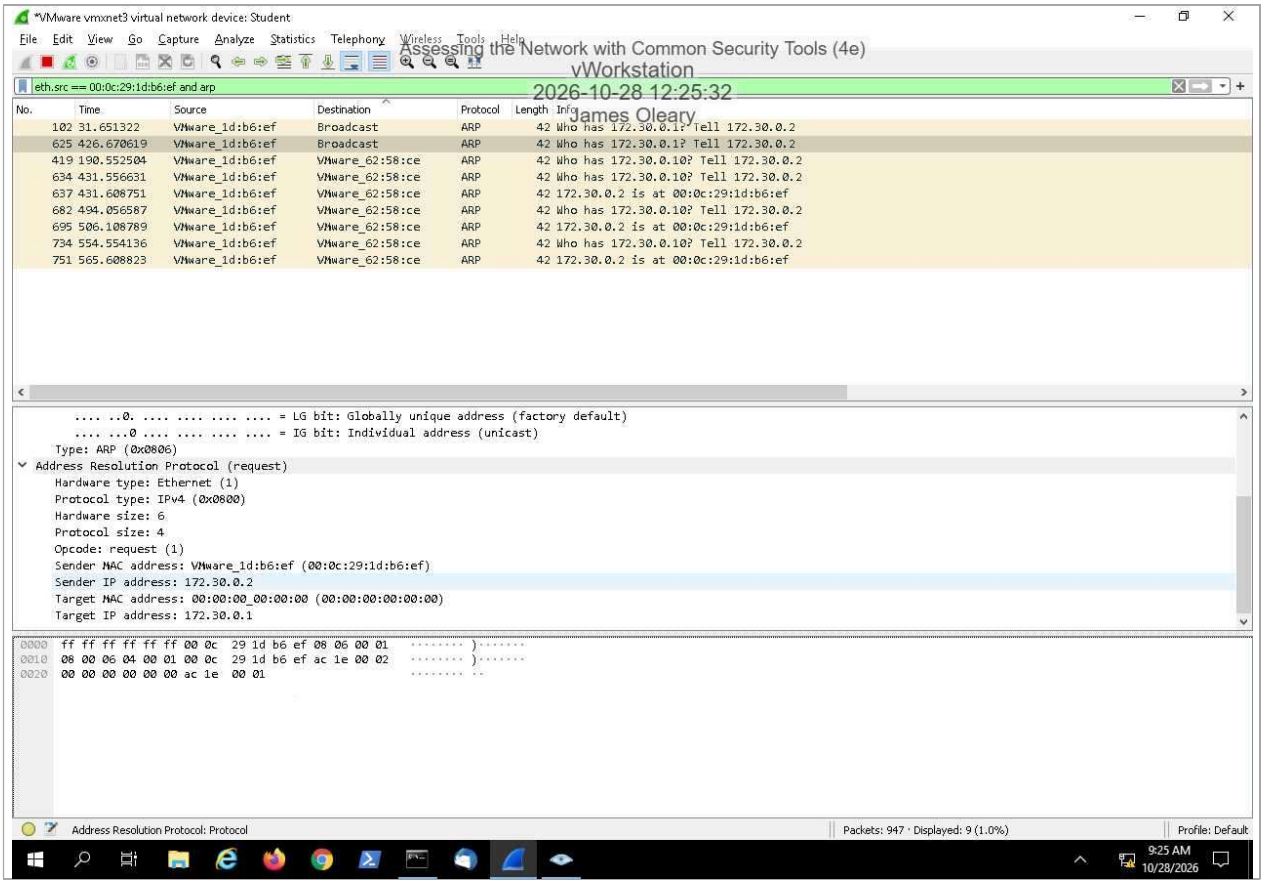
Required Screenshot 4: ICMP-Filtered Wireshark Packets



Note. The icmp display filter isolates packets 59 and 60: an echo request from vWorkstation (172.30.0.2) to TargetWindows01 (172.30.0.10) and the matching echo reply.

Figure 5

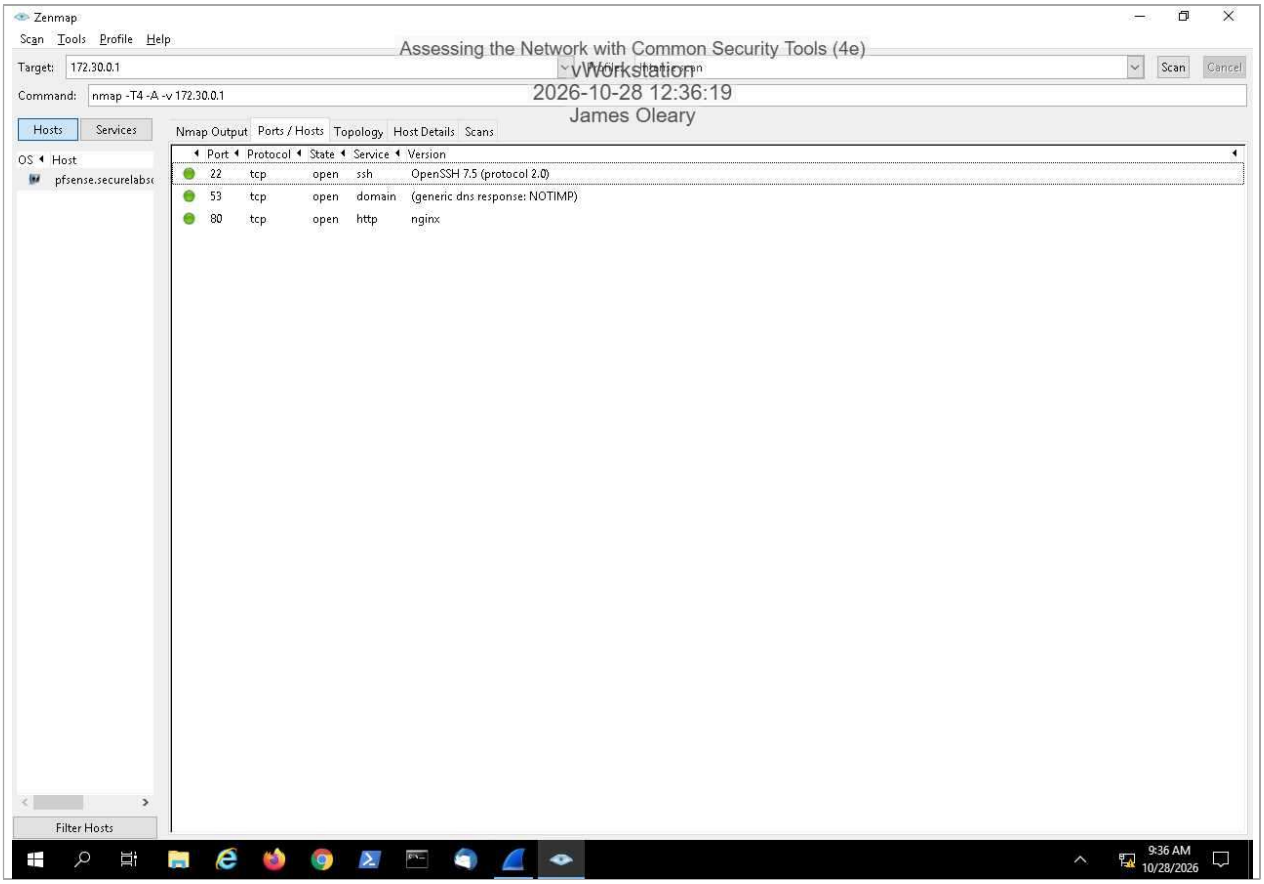
Required Screenshot 5: ARP-Filtered Wireshark Packets From vWorkstation to the Firewall



Note. Filtered with eth.src == 00:0c:29:1d:b6:ef and arp, packet 625 is the broadcast ARP request in which vWorkstation (172.30.0.2) asks for the MAC address of the firewall's internal interface (172.30.0.1).

Figure 6

Required Screenshot 6: Zenmap Ports/Hosts Tab



Note. The Intense scan (nmap -T4 -A -v 172.30.0.1) found three open TCP ports on the pfSense internal interface: 22 (SSH, OpenSSH 7.5), 53 (DNS), and 80 (HTTP, nginx).

Written Entry 1: Regular Scan Compared With the Ping Scan

For the ping scan the icmp filter showed no traffic to or from the firewall. Due to the fact Zenmap ran with admin permissions on the same subnet, Nmap checked to see if 172.30.0.1 was up by using an ARP packet instead. The regular scan started off in the same way with another ARP request but following that it went for port scanning, which uses TCP. The icmp filter didn't show packets involving the firewall, while the ARP filter showed a few. The regular scan seemed much more like the ping scan that expected. The only difference is probing common TCP ports.

Written Entry 2: Intense Scan Compared With the Ping Scan

The intense scan was much different compared to the ping scan. I noticed more icmp packets and ARP packets in the packet capture compared to when I did the ping scan. However most of what made intense scan much different was the things it did on top of icmp and ARP like service discovery and testing base scripts against a target.

Research Question 1: Security Measures That Protect a WAN

Identify and explain at least three security measures that protect a Wide Area Network (WAN). For each measure, describe (a) what it does, (b) which threat or attack surface it addresses, and (c) where it would be deployed in a WAN architecture (e.g., perimeter, host, link, or management plane).

A WAN is only as secure as the devices that connect it to the outside world, so I would secure it with 3 different controls so that there is defense in depth. The first control would be to harden the device and ensure they are always up to date. Turning off ports or services that a firewall, router, and VPN gateway don't need and installing vendor patches as quickly as possible (Australian Signals Directorate, 2025). Exploitation of known flaws in internet facing equipment is one of the most common ways for attackers to breach a network. In Verizon's 2025 breach report, they reported exploitation of edge devices and VPNs surged from 3% to 22% (Poireault, 2025). This control lives on the edge of a network. In the lab a Zenmap intense scan revealed pfsense running SSH, DNS, and HTTP on its internal interface. All of those should have a clear reason for why they are open. The second control is SASE with a zero-trust network access approach. Instead of giving remote workers access to a network through a VPN, SASE encrypts their traffic and grants access based on user identity markers like device and location (Gillis & Yasar, 2024). SASE fights eavesdropping on WAN connections and helps stop stolen accounts from spreading. My final control would be phishing-resistant MFA for admin logins (Australian Signals Directorate, 2025). A stolen admin password should not be enough to take over a router. This protects the management angle, where one compromise can affect the whole WAN.

References

Australian Signals Directorate. (2025). *Mitigation strategies for edge devices: Executive guidance*. https://media.defense.gov/2025/Feb/03/2003636713/-1/-1/0/CSI_MITIGATION-STRATEGIES-FOR-EDGE-DEVICES-EXECUTIVE-GUIDANCE.PDF

Gillis, A. S., & Yasar, K. (2024, June 13). *What is secure access service edge (SASE)?* TechTarget. https://www.techtarget.com/it-infrastructure/definition/secure-access-service-edge-SASE

Poireault, K. (2025, April 23). Verizon's DBIR reveals 34% jump in vulnerability exploitation. *Infosecurity Magazine*. https://www.infosecurity-magazine.com/news/verizon-dbir-jump-vulnerability/